

Total No. of Questions : 4]

SEAT No. :

PF205

[Total No. of Pages : 2

APR-26/SE/Insem-251

S.E. (Electronics/E & TC/Electronics & Computer Engg./Electronics

Engg. (VLSI Design & Technology)/Electronics &

Communication(Adv. Comu. Tech.))

SIGNALS & SYSTEMS

(2019 Pattern) (Semester-IV) (204191) (Insem)

Time : 1 Hour]

[Max. Marks : 30

Instructions to the candidates:

- 1) Answer Q.1 or Q.2 and Q.3 or Q.4.
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Figures to the right indicate full marks.
- 4) Use of electronic pocket calculator is allowed.
- 5) Assume suitable data if necessary.

Q1) a) Determine whether the signal is periodic, if it is periodic, find its period:
determine fundamental period. [5]

$$x(t) = 2\cos 100\pi t + 5\sin 50 t$$

b) Find Even and Odd components of the following signal. [5]

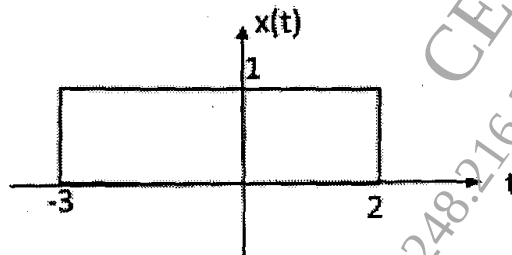
$$x(t) = \cos(t) + \sin(t) + \sin(t)\cos(t)$$

c) Determine whether the system described by $y(t) = 10 x(t) + 5$ [5]

- i) Memory less
- ii) Time invariant
- iii) Linear
- iv) Causal
- v) Stable

OR

Q2) a) For the signal shown in fig $x(t)$, perform the operation and draw $x(-t-2)$. [6]



P.T.O.

- b) Determine the following signal is Energy or Power signal & find the corresponding energy/power [5]

$$x(t) = e^{-5t} u(t)$$

- c) Evaluate the following integral using the property of impulse signal. [4]

$$\int_{-\infty}^{+\infty} e^{-5t^2} \delta(t-4) dt$$

- Q3)** a) Compute the convolution integral by graphical method and sketch the output for the following signal. [6]

$$x(t) = u(t+1) \text{ and } h(t) = u(t-2)$$

- b) An LTI system has impulse response $h(t) = u(t+1)$. [4]

Determine whether the system is causal, stable and memoryless.

- c) State and prove properties of convolution integral. [5]

OR

- Q4)** a) Compute the convolution sum of following sequence using graphical method. [6]

$$x[n] = \{1, 2, 1, 2\} \text{ and } h[n] = \{1, 1, 1\}$$

- b) Determine the Step response of the following system whose impulse response is $h(t) = e^{-5t} u(t)$. [4]

- c) Consider the interconnections of the systems as shown in fig. Let the impulse response be specified as [5]

$$h_1(t) = \delta(t), h_2(t) = \delta(t-1), h_3(t) = \delta(t+1), h_4(t) = u(t)$$

Obtain the overall impulse response of the system

